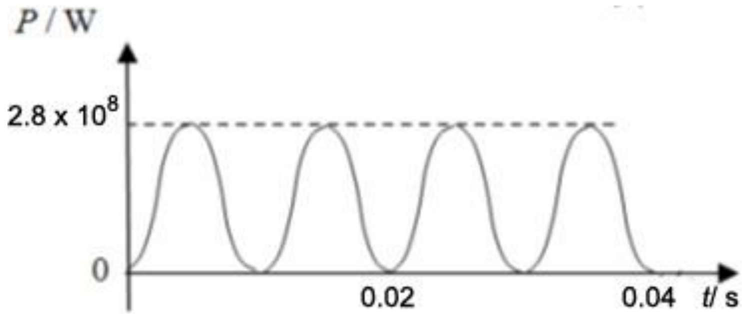
4 (a) Use P = VI

1. 2.8×10^8 (W) A1
2. 0.0 W A1

(b)



A $\sin^2$ graph with correct curvature drawn - 1 mark B1

Peak power at t = 0.005, 0.015, 0.025 s etc and Zero power at t = 0.010, 0.020, 0.030 s etc B1

(c) DC power = 800x350k = 2.8×10^8 W M1

AC average power = ½ peak power = 1.4×10^8 W, M1

so less average power than DC

(d) M1

V_s = 350k/20 = 17.5 kV

Ideal transformer, so power is unchanged,

$$2.8 \times 10^8 = 17.5\ \text{kV}\times I_s$$

secondary peak current = I_s = 16 kA

B1

The secondary peak current by a factor of 20, or increased to 16 kA.

5 (a) The **magnitude** of the electromotive force (e.m.f.) induced **in a** B1